

Quiz (Carolina Reaper)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -4$
(A) $(-2, 0)$
(B) $(-1, -1)$
(C) $(0, -2)$
(D) $(0, 0)$
(E) $(2, 0)$
- Q2.** What is the solution to the system: $y = x^2 - 4$, $y = -x^2 + 4$?
(A) $(0, 0)$
(B) $(0, 4)$
(C) $(2, 0)$
(D) $(-2, 0)$
(E) No solution
- Q3.** Graph the system: $y = x^2 - 2$, $y = x^2 + 2$. What is the number of intersection points?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** Solve the system: $x^2 + y^2 = 25$, $x - y = 3$
(A) $(4, -1)$
(B) $(-4, 1)$
(C) $(1, 4)$
(D) $(-1, -4)$
(E) $(2, 3)$
- Q5.** Solve the system: $y = x^2 + 1$, $y = -x^2 - 1$
(A) $(0, 1)$
(B) $(0, -1)$
(C) $(1, 0)$
(D) $(-1, 0)$
(E) No solution
- Q6.** Graph the system: $x^2 + y^2 = 16$, $x^2 - y^2 = 0$. What is the number of intersection points?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 0
- Q7.** Solve the system: $y = 2x^2 - 3$, $y = x^2 + 1$
(A) $(1, -1)$
(B) $(-1, 1)$
(C) $(0, -3)$
(D) $(2, 1)$
(E) $(-2, 5)$
- Q8.** What is the solution to the system: $y = x^2 - 2x + 1$, $y = -(x - 1)^2$?
(A) $(1, 0)$
(B) $(0, -1)$
(C) $(2, 1)$
(D) $(-2, 1)$
(E) No solution

Open Ended Questions (FRQ)

Q9. Solve the system of equations: $y = x^2 + 2x - 6$, $y = -x^2 - 2x - 2$. Show all steps and box the final answer.

Q10. Graph the system of equations: $x^2 + y^2 = 9$, $y = x^2 - 4$. Find the intersection points. Show all steps and box the final answer.

Chef's Tips

- Check for any algebraic or graphical errors
- Make sure students show all steps for free response questions
- Emphasize the importance of accurate graphing